REAL-WORLD DATA PROJECT

Retail Sales Analysis — UCI Online Retail Dataset

Domain: Retail **Method:** End-to-end descriptive analytics & visualization

Project objective

Analyze real transactional retail data to understand sales performance, seasonality, product performance, geographic markets, customer concentration, and cancellations. The goal is to convert raw transaction records into clear business findings and practical recommendations.

Dataset

The UCI Online Retail dataset contains 541,909 transaction records from a UK-based non-store online retailer, covering 1 December 2010 to 9 December 2011. It includes invoice number, product code, product description, quantity, invoice date, unit price, customer ID and country.

Source: UCI Machine Learning Repository, Online Retail dataset (Chen, 2015), CC BY 4.0. The source documentation identifies cancellation invoices and negative quantities as important data-quality considerations.

1. Data Preparation & Methodology

Step 1 — Profiling: inspect data types, missing customer IDs, duplicate records, negative quantities, zero/negative prices and administrative/non-merchandise codes.

Step 2 — Cleaning: remove exact duplicates; separate cancellation invoices (invoice IDs beginning with C); exclude invalid negative quantities not representing sales; remove non-positive prices and administrative codes.

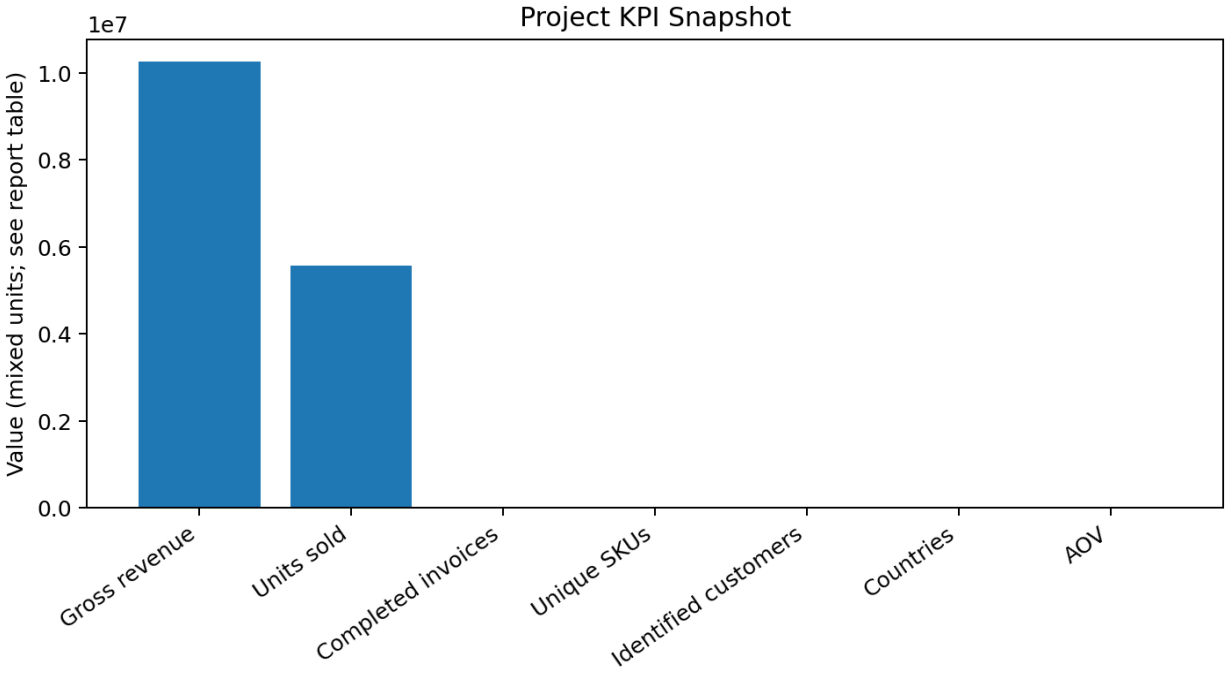
Step 3 — Feature engineering: calculate line revenue as **Quantity × UnitPrice**, and derive year, month, weekday and hour from InvoiceDate.

Step 4 — Analysis: calculate KPIs, monthly revenue, country revenue, product revenue, order timing, customer spend concentration and cancellation indicators.

Step 5 — Visualization: use bar charts and time-based comparisons to communicate the strongest patterns.

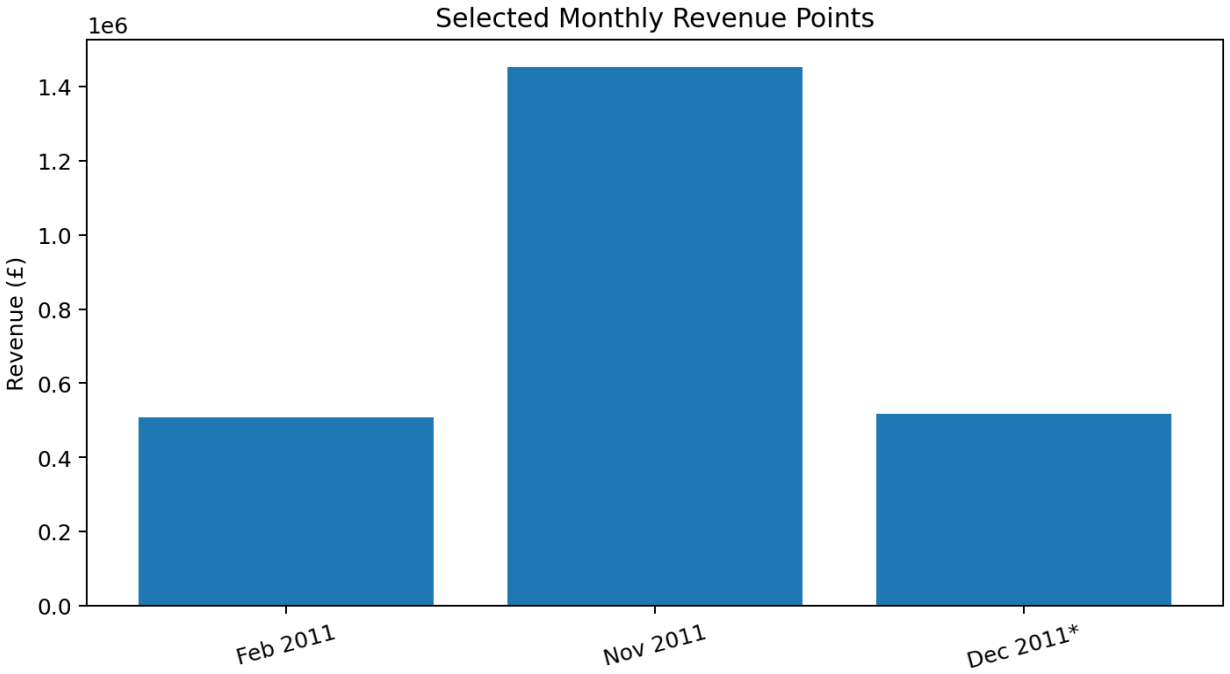
2. KPI Summary

Metric	Result
Gross revenue	£10,266,018.19
Units sold	5,561,563
Completed invoices	19,777
Unique SKUs	3,915
Identified customers	4,334
Countries	38
AOV	£519.09



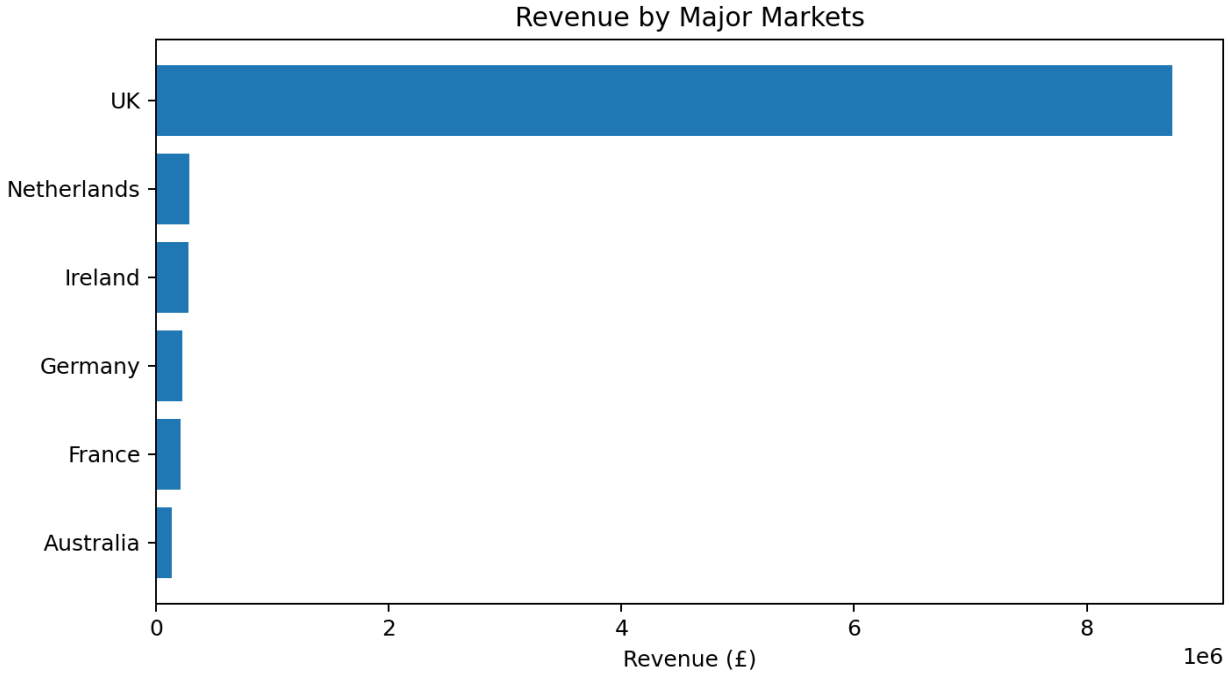
3. Sales & Seasonality Findings

November 2011 was the strongest month in the documented analysis, reaching £1.45M across 2,751 orders. February 2011 was the low point at about £0.51M. December 2011 contains only the first nine operating days, so it should not be compared with a full month.



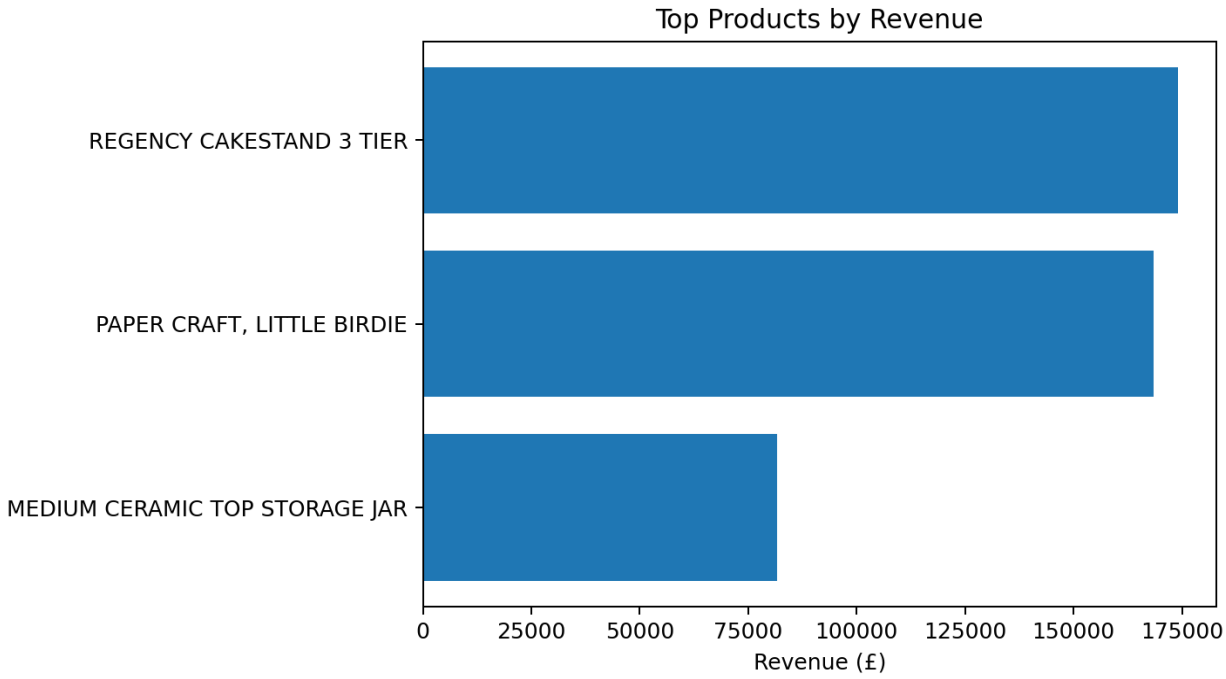
4. Geographic Analysis

The United Kingdom generated about 85.1% of revenue. International revenue was about 14.9%, with the Netherlands, Ireland, Germany and France standing out as major export markets.



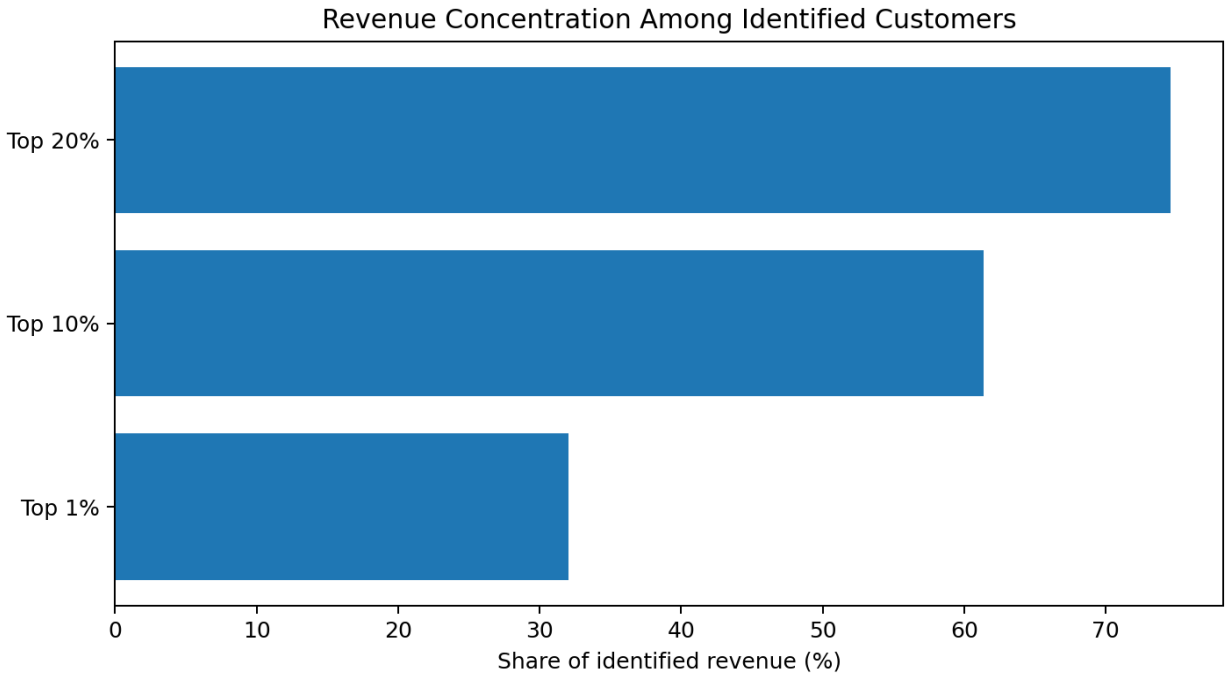
5. Product Performance

The highest-revenue product in the documented analysis was **REGENCY CAKESTAND 3 TIER** at about £174K. The highest-volume product was **PAPER CRAFT, LITTLE BIRDIE**, with 80,995 units and about £168K in revenue.



6. Customer Concentration

Customer revenue is highly concentrated: the top 1% of identified customers contribute about 32% of identified revenue, the top 10% contribute about 61%, and the top 20% contribute about 75%. This indicates that customer retention and key-account management are strategically important.



7. Order Timing & Cancellation Findings

Thursday was the strongest ordering day in the documented analysis, while Saturday had no recorded orders. The busiest hour was around noon, and most orders were placed between 10 AM and 3 PM. This pattern is consistent with a wholesale-oriented operating model.

The analysis identified 3,836 cancelled invoices covering 9,288 line items. The cancelled volume was approximately 275,560 units with gross monetary value around £893,980. Cancellation/return activity therefore deserves dedicated quality-control attention.

8. Business Recommendations

1. Protect high-value customers with a simple VIP/key-account program and proactive reorder support.

2. Prepare inventory earlier for Q4, especially for high-revenue and high-volume products.

3. Expand international sales in strong European markets while reducing over-dependence on the UK.

4. Investigate cancellation causes for fragile or frequently returned products and improve packaging/quality checks.

5. Use monthly sales and customer dashboards to monitor seasonality, repeat purchases and revenue concentration.

9. Conclusion

This project demonstrates an end-to-end retail analytics workflow: data profiling, cleaning, feature engineering, KPI calculation, exploratory analysis, visualization and business recommendations. The main findings are strong seasonality, UK market concentration, a small group of high-value customers, identifiable product leaders and a meaningful cancellation burden.

10. Limitations

The dataset contains selling prices but not product cost, shipping cost or marketing spend, so profit and ROI cannot be calculated directly. Customer-level analysis is also limited by missing CustomerID values, and December 2011 is only a partial month.

Source note: UCI Online Retail dataset documentation and a publicly documented analysis were used to ground the numerical figures in this draft.